Project Report: ROI Category Prediction for Marketing Campaigns (Healthcare)

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1. Data Understanding Report

1.1 Problem Statement

The given dataset containing below attributes for healthcare promotional data for social campaigns:

Input Variables:

1. campaign details

2. planned metrics

3. historical metrics

Output Variable:

1. roi_category

Predict roi_category (High / Medium / Low) for a campaign before it is launched.

Objective:

Predict whether a marketing campaign will achieve High, Medium, or Low ROI before launch, using campaign configuration and historical patterns.

Help marketing teams decide:

- Which campaigns to launch
- Reduce wasted spend
- Improve ROI planning

1.2 Business Context

In marketing campaigns, organizations invest significant budgets without certainty of returns.

A predictive system can:

- Identify high-performing campaigns in advance
- Optimize budget allocation

- Reduce financial risk
- Improve marketing efficiency

1.3 Objective

- Predict ROI category (Classification problem)
- Handle class imbalance (High ROI is minority)
- Avoid data leakage (use only pre-campaign features)
- Improve detection of high ROI campaigns
- Build an interpretable and practical model

1.4 Target Variable

- **roi_category**
  - 0 → Low ROI
  - 1 → Medium ROI
  - 2 → High ROI

1.5 Input Variables

Selected features include:

- Geography (country)
- Channel (Social, Email, etc.)
- Outlet (Facebook, Google, etc.)
- Campaign type
- Cost
- Planned impressions & clicks
- Planned CTR
- Historical CTR
- Conversion rate
- Budget utilization
- Seasonality index
- Audience reach score
- Engagement expectation
- Month

1.6 Initial Data Checks

Performed:

- df.shape → Dataset size (~4200 rows)
- df.info() → Data types
- df.describe() → Statistical summary

Observations:

- Dataset contains both categorical and numerical features
- Few missing values present
- Duplicate rows present

1.7 Data Cleaning Summary

Missing Values:

- Handled using **mean imputation**

Duplicates:

- Removed duplicate rows

Outliers:

- Treated using IQR/Z-score method (controlled, no major data loss)

Columns Removed:

- Dropped non-informative columns like data_source_id

1.8 Feature Engineering

Date Handling:

- Extracted **month** from campaign date

Encoding Preparation:

- Converted categorical features to numerical

1.9 Exploratory Data Analysis

Key Observations:

- High ROI campaigns are fewer (class imbalance)
- Most campaigns fall under Low/Medium ROI
- Weak correlation between features and target
- Planned clicks strongly related to impressions
- ROI depends on multiple interacting factors

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2. Feature Engineering Documentation

2.1 Encoding

- **Label Encoding** → for ordinal/categorical features
- **One-Hot Encoding** → for nominal features like:
  - Geography
  - Channel
  - Outlet

2.2 Feature Selection

To ensure realistic prediction:

Removed:

- Any feature directly derived from ROI (no leakage)

Kept:

- Only **pre-campaign features**

2.3 Final Feature Set

- Input: X (all encoded features)
- Target: y (roi_category)

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3. Model Development and Evaluation

3.1 Dataset Splitting

- Train: 80%
- Test: 20%

3.2 Models Implemented

1. Random Forest Classifier
2. Logistic Regression (Final Model)

3.3 Model Training

Models trained using:

```
model.fit(X_train, y_train)
```

3.4 Evaluation Metrics

- Accuracy
- Precision
- Recall
- F1-score (important for imbalance)

3.5 Results Summary

Final Model: Logistic Regression with class balancing

- Accuracy: **77%**
- Balanced performance across classes
- Significant improvement in High ROI detection

3.6 Confusion Matrix Insights

- Model performs well for Low and Medium ROI
- High ROI detection improved significantly
- Some false positives present (acceptable trade-off)

3.7 Model Selection

Selected Model: Logistic Regression

Reason:

- Handles class imbalance using class_weight
- Simple and interpretable
- Performs comparably to complex models
- Better generalization

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4. Insights and Recommendations

4.1 Key Insights

- High ROI campaigns are minority but critical
- Campaign planning features influence ROI
- Budget alone does not guarantee success
- Engagement and historical metrics are important

4.2 Top Contributing Features

- Planned clicks
- Planned CTR
- Historical CTR
- Conversion rate
- Budget utilization

4.3 Recommendations

- Focus on campaigns with strong engagement signals
- Improve planning for low-performing channels
- Use predictions before launching campaigns
- Monitor high-risk campaigns

4.4 Model Usage

- Input: Campaign planning data
- Output: ROI category (Low / Medium / High)

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5. Conclusion

The project successfully developed a machine learning model to predict ROI categories for marketing campaigns using pre-campaign data.

The Logistic Regression model with class balancing provided the best performance, especially in improving detection of high ROI campaigns.

This solution enables organizations to make data-driven decisions, optimize budgets, and improve campaign effectiveness.